CHAPTER EXERCISE: Answers for this chapter start on page 317.

A calculator should NOT be used on the following questions.

1

The expression $\frac{4x}{x-2}$ is equal to which of the following?

A) -2

B) $-\frac{8}{x-2}+4$

C) $\frac{8}{x-2}+4$

D) $4-2x$

2

If the expression $\frac{6x^2+5x+2}{2x+1}$ is written in the form $\frac{1}{2x+1}+Q$, what is Q in terms of x ?

A) $3x-1$

B) $3x+1$

C) $6x^2+3x+1$

D) $6x^2+5x+1$

3

The expression $4x^2+5$ can be written as $A(2x-1)+R$, where A is an expression in terms of x and R is a constant. What is the value of R ?

4

x	$g(x)$
-3	2
-2	3
0	-4
1	-3
3	6

The function g is defined by a polynomial. The table above shows some values of x and $g(x)$. What is the remainder when $g(x)$ is divided by $x+3$?

A) -2

B) 1

C) 2

D) 6

5

$$2z^3-kxz^2+5xz+2x-2$$

In the polynomial above, k is a constant. If $z-1$ is a factor of the polynomial above, what is the value of k ?

6

What is the remainder when x^2+2x+1 is divided by $x+4$?

A calculator is allowed on the following questions.

7

When $3x^2 - 8x - 4$ is divided by $3x - 2$, the result can be expressed as $A - \frac{8}{3x-2}$. What is A in terms of x ?

A) $x - 4$
B) $x - 2$
C) $x + 2$
D) $x + 4$

8

The expression $2x^2 - 4x - 3$ can be written as $A(x + 1) + B$, where B is a constant. What is A in terms of x ?

A) $2x + 6$
B) $2x + 2$
C) $2x - 2$
D) $2x - 6$

9

The expression $x^2 + 4x - 9$ can be written as $(ax + b)(x - 2) + c$, where a, b, and c are constants. What is the value of $a + b + c$?

A) -2
B) 3
C) 7
D) 10

10

For a polynomial $p(x)$, $p(2) = 0$. Which of the following must be true about $p(x)$?

A) $2x$ is a factor of $p(x)$.
B) $2x - 2$ is a factor of $p(x)$.
C) $x - 2$ is a factor of $p(x)$.
D) $x + 2$ is a factor of $p(x)$.

11

If $p(x) = x^3 + x^2 - 5x + 3$, then $p(x)$ is divisible by which of the following?

I. $x - 2$
II. $x - 1$
III. $x + 3$

A) I and II only
B) I and III only
C) II and III only
D) I, II, and III

12

If the polynomial $p(x)$ is divisible by $x - 2$, which of the following could be $p(x)$?

A) $p(x) = -x^2 + 5x - 14$
B) $p(x) = x^2 - 6x - 2$
C) $p(x) = 2x^2 + x - 8$
D) $p(x) = 3x^2 - 2x - 8$

13

If $x - 1$ and $x + 1$ are both factors of the polynomial $ax^4 + bx^3 - 3x^2 + 5x$ and a and b are constants, what is the value of a ?

A) -3
B) 1
C) 3
D) 5

14

For a polynomial $p(x)$, $p\left(\frac{1}{3}\right) = 0$. Which of the following must be a factor of $p(x)$?

A) $3x - 1$
B) $3x + 1$
C) $x - 3$
D) $x + 3$

Chapter 18: Synthetic Division

CHAPTER EXERCISE:

1. $\boxed{C}$

$$\begin{array}{r|l} & 4 \\ \hline x-2 & 4x \\ & \underline{4x - 8} \\ & 8 \end{array}$$

This result can be expressed as $4 + \dfrac{8}{x-2}$

2. $\boxed{B}$

$$\begin{array}{r|l} & 3x + 1 \\ \hline 2x+1 & 6x^2 + 5x + 2 \\ & \underline{6x^2 + 3x} \\ & 2x + 2 \\ & \underline{2x + 1} \\ & 1 \end{array}$$

This result can be expressed as

$3x + 1 + \dfrac{1}{2x+1}$, from which $Q = 3x + 1$.

3. $\boxed{6}$ This question is asking you to divide the expression by $2x - 1$ and write the result in the form of
Dividend = Quotient × Divisor + Remainder.

$$\begin{array}{r|l} & 2x + 1 \\ \hline 2x-1 & 4x^2 + 5 \\ & \underline{4x^2 - 2x} \\ & 2x + 5 \\ & \underline{2x - 1} \\ & 6 \end{array}$$

Therefore, $4x^2 + 5 = (2x + 1)(2x - 1) + 6$.
$R = 6$.

4. $\boxed{C}$ Using the remainder theorem, the remainder when $g(x)$ is divided by $x + 3$ is equal to $g(-3) = 2$.

5. $\boxed{7}$ $z - 1$ is a factor only if the polynomial yields 0 when $z = 1$ (the remainder theorem). Therefore, we can set up an equation.

$$\begin{aligned} 2(1)^3 - k(1)^2 + 5(1) + 2 - 2 &= 0 \\ 2 - k + 5 + 2 - 2 &= 0 \\ -k + 7 &= 0 \end{aligned}$$

From here, we can see that $k = 7$.

6. $\boxed{9}$ By the remainder theorem, the remainder is $(-4)^2 + 2(-4) + 1 = 16 - 8 + 1 = 9$.

7. $\boxed{B}$

$$\begin{array}{r|l} & x - 2 \\ \hline 3x-2 & 3x^2 - 8x - 4 \\ & \underline{3x^2 - 2x} \\ & - 6x - 4 \\ & \underline{- 6x + 4} \\ & - 8 \end{array}$$

This result can be expressed as

$x - 2 - \dfrac{8}{3x-2}$, from which $A = x - 2$.

8. $\boxed{D}$ This question is asking you to divide the expression by $x + 1$ and write the result in the form of
Dividend = Quotient × Divisor + Remainder.

$$\begin{array}{r|l} & 2x - 6 \\ \hline x+1 & 2x^2 - 4x - 3 \\ & \underline{2x^2 + 2x} \\ & - 6x - 3 \\ & \underline{- 6x - 6} \\ & 3 \end{array}$$

Therefore, $2x^2 - 4x - 3 = (2x - 6)(x + 1) + 3$.

9. [D] This question is asking you to divide the expression by $x-2$ and write the result in the form of
Dividend = Quotient × Divisor + Remainder,
where $ax+b$ is the quotient and c is the remainder.

$$\begin{array}{r|l} & x + 6 \\ \hline x-2 & x^2 + 4x - 9 \\ & x^2 - 2x \\ \hline & 6x - 9 \\ & 6x - 12 \\ \hline & 3 \end{array}$$

Therefore, $x^2+4x-9=(x+6)(x-2)+3$.
Finally, $a=1, b=6, c=3$, and $a+b+c=10$.

10. [C] Using the remainder theorem, $p(2)=0$ means that $x-2$ is a factor of $p(x)$.

11. [C] Use the remainder theorem to test each option for a remainder of 0.
$p(2)=2^3+2^2-5(2)+3=5$.
$p(1)=1^3+1^2-5(1)+3=0$.
$p(-3)=(-3)^3+(-3)^2-5(-3)+3=0$.
Therefore, $p(x)$ is divisible by $x-1$ and $x+3$.

12. [D] If $p(x)$ is divisible by $x-2$, then $p(2)$ must equal 0 (the remainder theorem). Testing each answer choice, only choice (D) results in 0 when $x=2$.

13. [C] Using the remainder theorem, we can set up a system of equations. When the polynomial is divided by $x-1$ or $x+1$, the remainder is 0, which means that if we let $p(x)$ denote the polynomial, $p(1)=0$ and $p(-1)=0$.

$$\begin{cases} a(1)^4+b(1)^3-3(1)^2+5(1) & =0 \\ a(-1)^4+b(-1)^3-3(-1)^2+5(-1) & =0 \end{cases}$$

$$\begin{cases} a+b-3+5 & =0 \\ a-b-3-5 & =0 \end{cases}$$

Adding the equations together,

$$\begin{aligned} 2a-6&=0 \\ a&=3 \end{aligned}$$

14. [A] From the remainder theorem, $3x-1$ must be a factor of $p(x)$ if $p\left(\frac{1}{3}\right)=0$.